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Hybrid learning strategies for multivariate time series forecasting of network quality metrics

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ABSTRACT

This work addresses the challenge of forecasting temporal metrics that characterize cellular traffic behavior. The ultimate goal is to provide network operators with a valuable tool for modeling mobile network traffic and optimizing connected resources. The idea is to estimate beforehand the temporal evolution of some Quality-of-Experience (QoE) and Quality-of-Service (QoS) metrics, which is helpful for accurately tuning the allocation of network resources. Remarkably, these metrics (expressed as time series) are typically correlated, and changes in one time series can affect others in a variety of ways and to different extents. For example, high network delay (a QoS-related metric) is associated with degradation in voice quality over time (a QoE-related metric). Accordingly, we address the problem of cellular traffic forecasting with correlated time series, proposing three innovative hybrid learning strategies designed by combining the advantages of two approaches: (*i*) a statistical approach, implemented through the Vector Autoregressive (VAR) model, which encodes each metric as a combination of past values of the same metric along with a combination of values of other related metrics, resulting in a *multivariate* structure; and (*ii*) an approach based on deep learning techniques (specifically, CNN, LSTM, and GRU) which operate on such a multivariate structure to perform the forecasting. The resulting performance demonstrates the benefits of the proposed hybrid schemes (VAR-CNN, VAR-LSTM, VAR-GRU) over their *pure* counterparts, with a significant reduction in forecasting errors. The network metrics were gathered in a real urban cellular environment, where the presence of exogenous factors (e.g., interferences, weather conditions, etc.) makes the forecasting assessment particularly challenging.

1. Introduction and contribution

Time series forecasting represents an important field of research which spans multiple domains including financial markets [1,2], biomedical signals [3,4], sentiment analysis [5,6], cloud provisiong [7,8], where it can be exploited to estimate future values of observations exhibiting a chronological order. In particular, in the mobile network management field, time series forecasting is helpful to estimate the trend of some time-variant metrics (both related to Quality of Experience — QoE and to Quality of Service — QoS) thus allowing mobile operators to optimize resource and infrastructure usage. The wireless channel (affected by unpredictable phenomena such as congestion, errors, etc.) has a great impact on the behavior of these metrics which, consequently, show a strong temporal correlation that we capture through a *multivariate* time series model. For instance, bad channel conditions cause a concurrent degradation of voice quality (a QoE-related metric) and of network delay (a QoS-related metric).

Until a few years ago, classic statistical approaches dominated the field of time series forecasting through the usage of Autoregressive (AR), Integrated (I), Moving Average (MA) models and their combinations (e.g., ARMA, ARIMA). Such models allow full control of the analytical structure of a time series but can suffer from the presence of significant irregularities [9]. In contrast, deep learning techniques seem to be effective in dealing with time series forecasting, due to the implementation of densely connected structures suitable to capture complex and non-linear behaviors. A drawback is that, by definition, deep learning techniques adopt a black-box approach, thus the values of hyperparameters have to be chosen in advance [10].

Aware of the different purposes of statistical and deep learning techniques, in this work we combine the *pros* of such two approaches: the capability of statistical methods to capture the underlying time series structure, with the power of generalization offered by deep learning during the forecasting stage. The result is a set of hybrid

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learning strategies that we have profitably employed to forecast the behavior of some QoE/QoS network metrics (some of them seldom analyzed in literature) gathered during a number of mobile real-time traffic experiments [11] including: the Mean Opinion Score (MOS) which quantifies the perceived quality of a conversation, the bandwidth occupied by a voice flow (BW), the Round Trip Delay (RTD) which is a measure of the network delay, the jitter (JIT) affecting the wireless channel, the De-Jittering Buffer (DJB) whose size varies according to the jitter, and the Signal-to-Noise Ratio (SNR) quantifying the signal power with respect to the noise power.

Among the possible statistical approaches, we adopt the Vector Autoregressive (VAR) model which allows to express a given time series as a linear combination of its past values along with past values of other (related) time series. Such a multivariate characterization is useful within the realm of network traffic modeling, since the involved metrics typically are reciprocally influenced (for example, high RTD negatively impacts on the MOS).

The produced multivariate time series feeds three different deep-based schemes: a Convolutional Neural Network (CNN), a Long-Short Term Memory (LSTM) network, and a Gated Recurrent Unit (GRU) network. Even if traditionally used into the image processing domain, CNNs have been chosen for their ability to treat the input data as a 1-dimensional array (a time sequence of observations) and to extract from convolutional layers valuable information for the final forecast. In contrast, LSTM and GRU (belonging to the recurrent neural networks - RNNs - family) have been designed to handle the complexity of past-future dependencies, thus they represent a natural tool to manage time series.

The main contributions can be summarized as follows:

1. We propose three hybrid learning strategies (VAR-CNN, VAR-LSTM, VAR-GRU) to forecast time series traffic relying on a two-stage procedure: (*i*) a VAR model produces a multivariate time series structure along with estimates of crucial parameters, and (*ii*) three deep learning techniques (CNN, LSTM, GRU) perform the forecast. Performance results show the effectiveness of the proposed hybrid strategies if compared with their *pure* counterparts, since, with a negligible impact on computational times, the forecasting error in some cases is more than halved.
2. We deeply examine subtle interdependencies among time series through particular analyses such as the Forward Error Variance Decomposition (FEVD) which is useful to understand how some time series react when other (related) time series are subject to an exogenous change. In particular, this analysis explains the variations of the variables within the model.
3. From a real experimental campaign (realized across a real cellular LTE-A scenario) we extract and elaborate a set of QoE/QoS metrics characterizing the mobile cellular traffic. In order to apply learning-based techniques to metrics forecasting, we recast the time series format of QoE/QoS metrics into a supervised learning framework through the sliding window mechanism.

The remainder of the paper is organized as follows: In Section 2 we discuss the differences between our proposal and the affine technical literature; Section 3 describes the experimental testbed implemented to analyze network metrics gathered from a campaign of data collection; In Section 4 we provide details about the multivariate time series construction along with some statistical techniques and methods amenable to explore hidden relationships among time series; Section 5 provides details about deep learning techniques employed to perform the forecasting stage; Section 6 is devoted to the description of results obtained by the proposed hybrid techniques along with performance discussion. Finally, Section 7 concludes the paper along with hints for future research.

2. Related work

Recently, the interest in time series prediction applied to the network traffic is grown also boosted by new virtualized networks which allow to allocate resources on the basis of an actual traffic load. In particular, both statistical and learning-based approaches have been proposed and examined across the technical literature [12–14].

We start to describe the former approaches typically adopted in the literature. For instance, authors in [15] propose a framework for the prediction of self-similar network traffic by using an AutoRegressive Moving Average (ARMA) model analyzing two metrics: the packet size and the interval between two consequent packets. An ARMA model has been also employed in [16] to predict P2P network traffic. In [17], authors exploit the AutoRegressive Integrated Moving Average (ARIMA) model along with a back-propagation neural network mechanism and a simulated annealing algorithm to forecast the network traffic. Moreover, some tests including stationarity and autocorrelation have been performed to evaluate the features of the time series under analysis. Authors in [18] employ the ARIMA model to perform LTE network traffic prediction, where the choice of model parameters has been made in accordance to the Box-Jenkins test. ARIMA models have also been used in combination with some learning-based techniques (e.g. LSTM) to achieve prediction of network traffic as shown in [19] and in [20]. A variant of ARIMA, useful to deal with a seasonal component of time series and called SARIMA, has been used in [21] and in [22] by exploiting the property of periodicity in mobile traffic. One drawback of the presented methods (ARMA/ARIMA/SARIMA) is that they are able to deal with univariate time series. In contrast, we employ the vector autoregressive models being interested in exploring the inter-dependencies among multiple time series, and exploiting such correlations to improve the quality of the forecasting.

We now discuss another part of the literature which has examined learning-based techniques (with a particular focus on deep learning approaches). The authors in [23] examine both CNNs and RNNs for traffic forecasting and find that the formers are able to extract geographical traffic features, whereas the latter are able to extract temporal traffic features. A framework called DeepTP is proposed in [24], and it is employed to forecast traffic demands from spatial-dependent and long-period cellular traffic. Such a framework is composed by a feature extractor for modeling spatial dependencies, and by a sequential module for modeling temporal variations. Techniques based on Long Short-Term Memory (LSTM) are presented in [25–30], and [31] to predict traffic data in: IoT networks, backbone networks, VoIP scenarios, CDNs, cellular networks, streaming platforms, and campus networks, respectively. An overview of deep learning models applied for multivariate time series classification in outdoor/indoor IoT environments has been proposed in [32]. One drawback of "pure" learning-based techniques is that they basically neglect the presence of a structure behind the time series. Another track of research into network traffic prediction is focusing on the graph neural networks [33–38]. One significant drawback of such methods is that they require precise knowledge of providers network topologies, which, in most cases, are kept confidential for privacy and commercial purposes.

In summary, statistical methods offer the advantage of providing a convenient analytical representation, but they often fail to effectively capture unpredictable irregularities in time series. Additionally, many works that introduce statistical methods for forecasting tend to utilize a univariate approach, treating each time series as disjoint from others. In contrast, deep learning methods are advantageous because they are independent of the underlying analytical model (if any), but they require significant computational resources, especially when tuning a large number of hyperparameters. In our proposal, we leverage the benefits of combining statistical and learning-based techniques for predicting network metrics. Specifically, our hybrid schemes utilize a multivariate statistical approach to construct the "skeleton" of a time series model (first stage), while employing deep learning for forecasting (second stage).

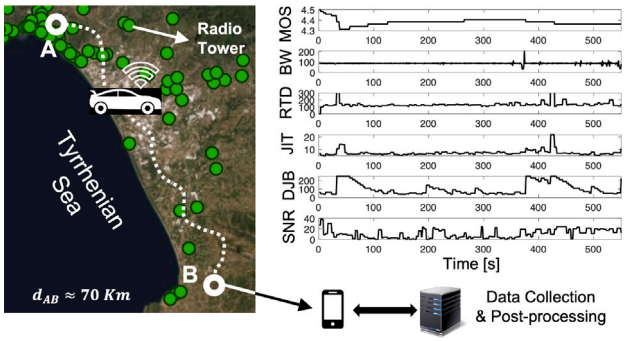

Fig. 1. Experimental testbed: data collection (cellular traffic) in a real environment, and time series arrangement.

3. Experimental design

Fig. 1 shows a simplified representation of carried experiments, where mobile (VoIP) traffic has been gathered and post-processed. Each experiment involves two mobile devices which are synchronized through the Network Time Protocol. The first device is on a car which starts to move from point A and covers a distance of about 70 Km up to point B (see the white dashed curve from A to B on the map in the left part of Fig. 1) with a 60 Km/h speed, on average.

The test area (close to the Tyrrhenian Sea — Southern Italy) is densely populated (about 2000 people/Km^2) with about 100 radio towers (green dots on the map) in LTE-Advanced technology [39]. In correspondence of point B are located: a fixed user (the second device), and a server in charge of processing the data traffic acquired during the communication between the two devices.

Both devices are equipped with Linphone, a soft-phone supporting the Real Time Control Protocol — Extended Reports (RTCP-XR) [40] allowing to track different metrics, including MOS and SNR, which are rarely taken into account across technical literature. In particular, RTCP-XR carries specific QoE/QoS information in a dedicated header part named VoIP metrics Report Block. The collected traffic is divided in patterns each one amounting to about 550 seconds and containing approximately 50,000 voice packets. The adopted audio codec is G.722 working at 7 kHz of sampling rate and 64 kb/s [41]. Finally, through customized filters, we extract and arrange in a time series format the following metrics:

Mean Opinion Score (MOS): it quantifies the perceived quality during a voice call through a pure number ranging between 1 (low perceived quality) and 5 (high perceived quality).

Bandwidth (BW): it provides information pertaining to the voice call bandwidth consumption (in kb/s). High BW values typically indicate an inefficient usage of channel resources.

Round Trip Delay (RTD): it measures the temporal interval (in ms) between a voice packet sent from a source and the *ack* message received from the destination. High RTD values are usually associated to a network congestion.

Jitter (JIT): it provides a measure (in ms) of the delay variability between consecutive voice packets. High JIT values imply distortion within voice communication.

De-Jittering Buffer (DJB): it measures (in packets converted in ms) the size of the buffer introduced to retain voice packets for some time so that the end user could not experience voice distortion. High DJB values are necessary in case of high network jitter.

Signal-to-Noise Ratio (SNR): it quantifies (in Decibel — dB) the objective quality of a communication channel, being defined as the ratio between the power of a signal and the power of the background noise. High SNR values indicate a good channel quality.

It is worth noting that, being in a real environment, the six collected times-series metrics reflect the real status of the wireless channel which is unavoidably affected by: weather condition, presence of interfering communications, time-varying load of network nodes, obstacles. Accordingly, as it will be evident in Fig. 9 further ahead, time series occasionally exhibit deviations from their expected trends, thus making the forecasting problem notably challenging.

4. Multivariate time series modeling and forecasting

In multivariate forecasting problems, the value of one time series is not only related to its predecessors in time but, in addition, it depends on past values of other related time series [42]. In practice, the current value of a time series can be easily represented by a *linear combination* of past values of the time series itself and past values of other time series.

The amount of "past" (denoted by p) is commonly known as *lag*, and the resulting VAR(p) model can be written as:

$$y_t = c + A_1 y_{t-1} + A_2 y_{t-2} + \cdots + A_p y_{t-p} + \epsilon_t, \tag{1}$$

where: $y_t = (y_{1t}, y_{2t}, \dots, y_{Nt})^T$ is the vector of $N = 6$ time series (MOS, BW, RTD, JIT, DJB, SNR) at time t which depends on past values $(t-1, \dots, t-p)$, c is a vector of constants, A_k are the $N \times N$ matrices of coefficients $(k = 1, \dots, p)$ embedding the past contributions of all involved time series, and $\epsilon_t = (\epsilon_{1t}, \epsilon_{2t}, \dots, \epsilon_{Nt})^T$ is a vector of zero-mean errors (or residuals).

When analyzing VAR processes, it is useful to perform some preliminary analyses to understand if time series composing the process fulfill specific properties. In particular, we first perform some analyses useful to inspect valuable structural properties of time series, and then we investigate about the time series relationships.

4.1. Preliminary statistical analyses

Akaike Information Criterion. A natural question that arises is how to determine the optimal lag p to take into account in the VAR process: the idea is to choose the lag such that a measure of error forecast is minimized. At this aim we adopt the Akaike Information Criterion (AIC) [43] which estimates the relative amount of error in selecting a lag: the less AIC value in correspondence of a given lag, the higher the quality of forecast.

Precisely, the AIC value amounts to: $\log|\Sigma_\epsilon| + 2pN/L$, where $|\Sigma_\epsilon|$ is the determinant of the covariance matrix of residuals estimated through the Ordinary Least Squares method, and L is the time series length.

In the leftmost part of Fig. 2 we have summarized the two steps required to build a VAR model: first, AIC is applied (along with additional tests) to estimate the optimal lag p, and, then, the multivariate time series model is built by embodying the p past values and the estimated A_k matrices. The minimum value returned by applying AIC amounts to 11. This notwithstanding, also in accordance with credited literature [44], we decided to refine such a result through additional tests aimed to ensure uncorrelated residuals (see below).

Uncorrelation of Residuals. A desirable property of a VAR model is that the residuals are uncorrelated in order to have unbiased forecasting.

For this task, we employ the Breusch–Godfrey test [45] (B-G test, in brief) useful to discover a possible non-negligible autocorrelation in the residuals. We start from the VAR model whose lag is the one minimizing AIC, and then we test the serial correlation of the residuals through the B-G test. Such a test starts from the expression of the VAR model associated to the error ϵ_t (a.k.a. auxiliary model), namely

$$\epsilon_t = \Omega_1 \epsilon_{t-1} + \cdots + \Omega_s \epsilon_{t-s} + n_t, \tag{2}$$

being s the maximum lag of the auxiliary model, Ω_ξ $(\xi = 1, \dots, s)$ the pertinent matrices of coefficients, and n_t a white noise. The B-G test is a statistical hypothesis test formulated as follows:

$$\begin{cases} \mathcal{H}_0 : \Omega_1 = \cdots = \Omega_s = 0, \\ \mathcal{H}_a : \Omega_i \neq 0 \quad \text{for at least one } i \in \{1, \dots, s\}. \end{cases} \tag{3}$$

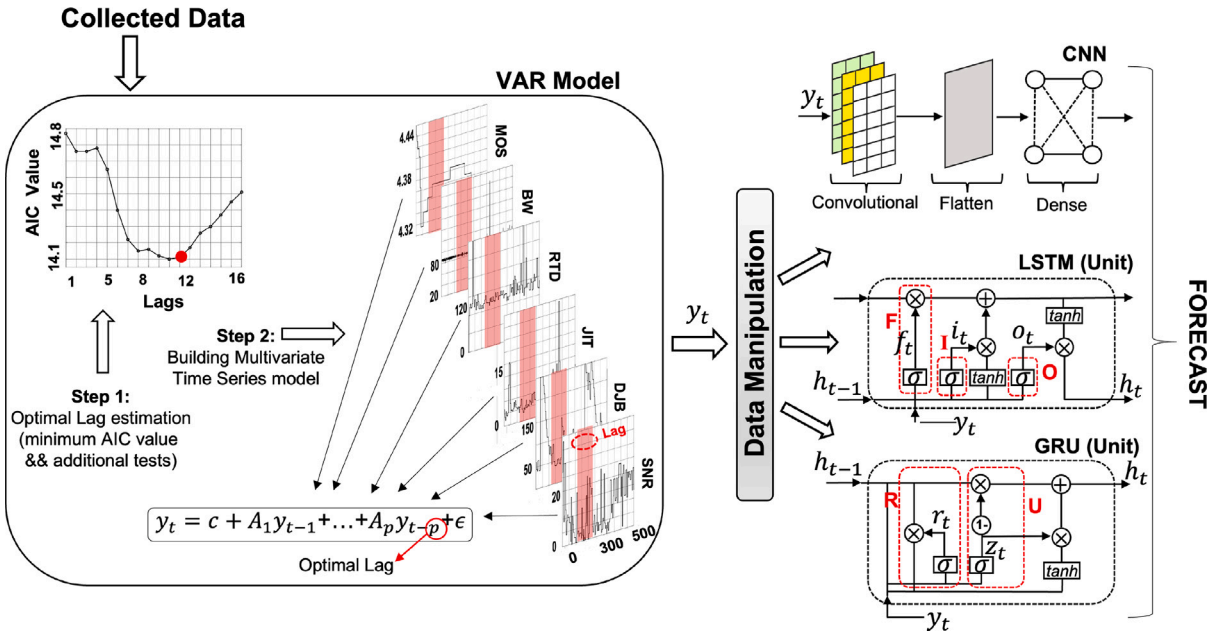

Fig. 2. Multivariate time series characterization of collected metrics through combination of VAR modeling (leftmost part) and deep-based forecasting techniques: CNN, LSTM, and GRU (rightmost part).

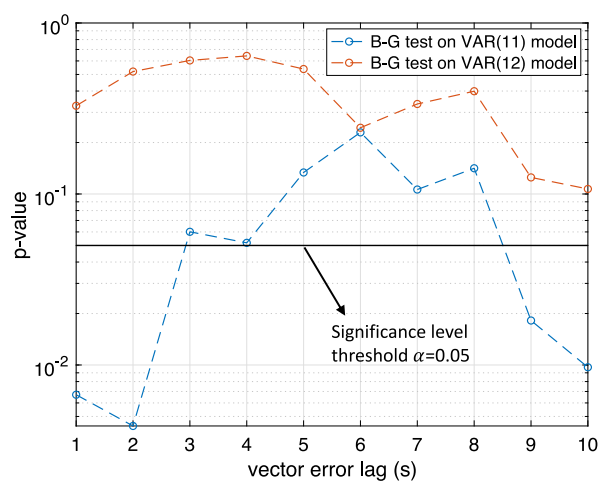

Fig. 3. Results of Breusch–Godfrey (B–G) tests (in terms of p-values) associated to VAR(11) and VAR(12) models.

The hypothesis $\mathcal{H}_0$ (or *null* hypothesis) implies the absence of a significant autocorrelation of residuals, whereas the hypothesis $\mathcal{H}_a$ (or *alternative* hypothesis) implies the presence of a non-negligible autocorrelation of residuals. To evaluate test (3), we calculate the p-value, namely the probability that quantifies the strength of the evidence against the null hypothesis $\mathcal{H}_0$. Pragmatically, if the p-value is less or equal than a given significance level α that we conventionally set to 0.05, we reject $\mathcal{H}_0$ with an error probability of 5% at most. We employ the B-G test both on the VAR(11) and on the VAR(12) models, being 12 the second optimal lag value returned by the minimization of the AIC metric. Results are presented in Fig. 3, where we show (on a semi-log scale) the p-value behavior for VAR(11) in blue and for VAR(12) in red in an interval of s chosen according to [46]. We see that, for $p = 12$, all the corresponding p-values are above the threshold $\alpha = 0.05$ (the black horizontal curve), thus we are more guaranteed that the VAR(12) model exhibits negligible values of the autocorrelation function of residuals ϵ_t for $t > 0$, with respect to the VAR(11) model.

Once pinpointed the best lag value (12 in our case), it is useful to evaluate the autocorrelation $R_\epsilon(t_i, t_j) = \mathbb{E}[\epsilon_{it} \cdot \epsilon_{jt}]$, and to verify that its values are as small as possible (ideally, 0) for $i \neq j$. Panel of Figs. 4 reports the autocorrelation of residuals for each time series as a function of the lag. In practice, we are interested that autocorrelation exhibits a very small value for $p = 12$ (the optimal) but, we verify that such a condition holds true for a broad range of lags spanning from 1 to 30. We also note that the residual correlation amounts obviously to 1 (the maximum value) at $p = 0$.

Stationarity. When employing VAR models for forecasting, it is important to ensure that the composing time series are stationary. Since time series can be interpreted as random processes, stationarity implies that mean and variance of such processes do not change over time. A common cause of non-stationarity is the presence of trend or seasonality within a time series, making its moments to be time-variant. To perform the stationarity check, we start from the stability condition which, in turn, implies the stationarity [42]. Formally, y_t is stable if

$$|I_N - A_1 z - \cdots - A_P z^p| \neq 0 \quad \text{for } |z| \leq 1, \tag{4}$$

where $|\cdot|$ is the determinant and I_N the N-order identity matrix.[1] Interestingly, condition (4) is equivalent to the condition that all eigenvalues of the companion matrix Φ have modulus less than 1 [46], being

$$\Phi = \begin{bmatrix} A_1 & A_2 & \ldots & A_{p-1} & A_p \\ I_N & 0 & \ldots & 0 & 0 \\ 0 & I_N & \ldots & 0 & 0 \\ \vdots & & \ddots & \vdots & \vdots \\ 0 & 0 & \ldots & I_N & 0 \end{bmatrix}. \tag{5}$$

The result of such a check is reported in Fig. 5 which confirms the stationarity hypothesis.

4.2. Metrics relationships

VAR model in (1) implicitly assumes some relationship among the involved time series. This notwithstanding, we have to employ other

[1] It is equivalent to say that the polynomial $|I_N - A_1 z - \cdots - A_P z^p|$ has no roots in and on the complex unit circle.

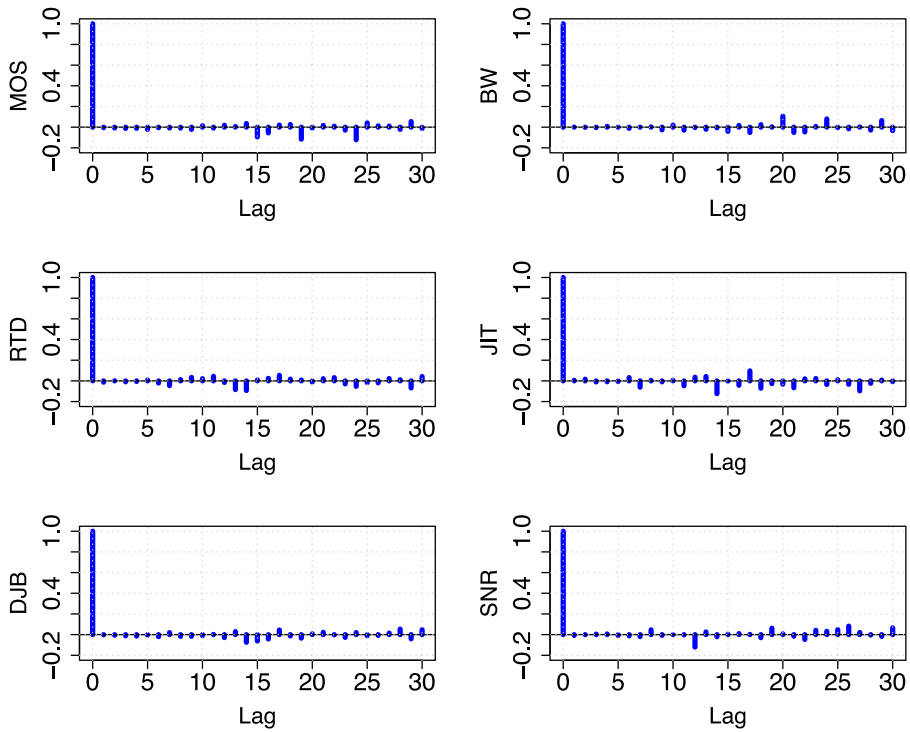

Fig. 4. Autocorrelation of Residuals.

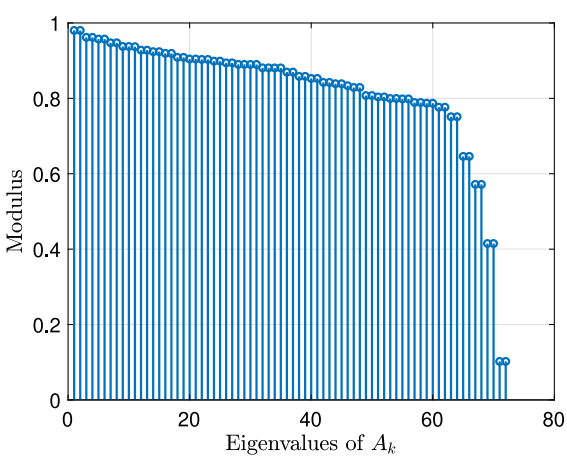

Fig. 5. Modulus of A_k eigenvalues which, being < 1, implies stability and then stationarity.

techniques to deeply examine the nature of such a relationship. One of the best methodologies for this kind of analysis relies on the Pearson correlation coefficient defined as:

$$\rho_{y_i,y_j} = \frac{E[(y_i - \mu_{y_i})(y_j - \mu_{y_j})]}{\sigma_{y_i}\sigma_{y_j}}, \tag{6}$$

where y_i, μ_{y_i}, and σ_{y_i} refer to the i-th times series, its mean value, and its standard deviation, respectively. We recall that the Pearson coefficient can assume a range of values spanning from -1 to 1: -1 indicates the maximum negative correlation between two time series, 0 indicates no correlation, and 1 indicates maximum positive correlation. For the sake of comfort, we have reported in the graphical pie matrix of Fig. 6 the result of such a correlation analysis. The fuller the pie in blue [red] in the (i, j) position, the higher the positive [negative] correlation between time series i and j. As expected, on the main diagonal of the pie matrix we see completely blue-colored filled pies indicating the maximum correlation between a time series with itself. As regards other correlations we can note some facts. First, the three delay-based time series (RTD, JIT, and DJB) are positively correlated: for instance, when jitter increases, also round trip delay and de-jitter buffer increase as correctly expected. MOS is weakly correlated with the SNR since a good voice quality perception is somehow connected with the quality of the underlying channel. Conversely, MOS is anti-correlated with RTD, JIT, and DJB since an increase of such metrics is connected with a deterioration of the voice quality perception. Finally, the bandwidth consumption seems to exhibit a weak correlation with other variables. For example, as BW increases, RTD slightly decreases (and vice versa) since the occupation of a wider bandwidth makes the round trip delay to be more restrained.

The correlation matrix in Fig. 6 can be regarded as a static snapshot highlighting the relationships among the time series belonging to the VAR(p) process. Conversely, it could be useful to examine how an external disturbance (often known as *shock* or *innovation*) in a time series can propagate and influence the remaining time series. One of the analyses allowing to examine such a situation is the forecast error variance decomposition (FEVD), used into econometrics to predict the volatility of the stock market [47]. Given f the forecast horizon, FEVD helps to understand what portion of the variance of the forecast error in forecasting y_{t+f} is due to a structural shock w_j.

To examine such a technique, it is useful to rewrite the process (1) according to the transformed Wold representation [48]:

$$y_t = c + \sum_{i=0}^{\infty} \Theta_i w_{t-i}, \tag{7}$$

where: $\Theta_i = \Phi_i P$, being P a lower triangular nonsingular matrix with positive diagonal elements derived according to the *Choleski* decomposition [49]. Moreover, $w_t = P^{-1}\epsilon_t$ is a white noise with covariance matrix $\Sigma_w = \mathbb{E}[w_t w_t^T] = I_N$. Since white noise errors w_t exhibit

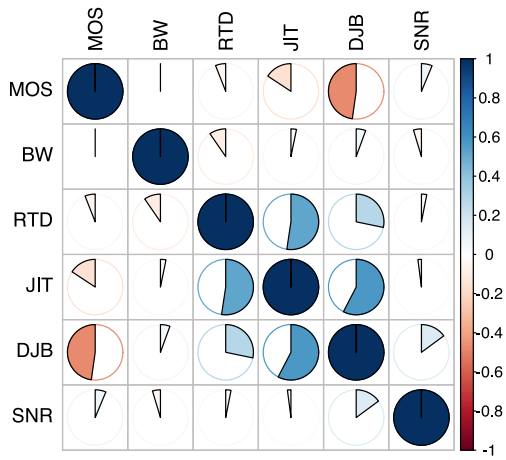

Fig. 6. Correlation Matrix.

uncorrelated components $w_{1t}, \dots, w_{Nt}$ with unit variance I_N, they are known as *orthogonal* residuals or innovations.

In view of (7), the f-step forecast error is:

$$y_{t+f} - y_t = \sum_{i=0}^{f-1} \Theta_i w_{t+f-i}. \tag{8}$$

Denoting by $\theta_{rs,i}$ the rs-th element of Θ_i, the f-step forecast error of the ℓ-th component of y_t can be expressed as:

$$\begin{aligned} y_{\ell,t+f} - y_{\ell,t} &= \sum_{i=0}^{f-1} \left(\theta_{\ell 1,i} w_{1,t+f-i} + \cdots + \theta_{\ell N,i} w_{N,t+f-i}\right) \\ &= \sum_{n=1}^{N} \left(\theta_{\ell n,0} w_{n,t+f} + \cdots + \theta_{\ell n,f-1} w_{n,t+1}\right), \end{aligned} \tag{9}$$

where the forecast error of the ℓ-th component is made of all innovations $w_{1t}, \dots, w_{Nt}$. Due to the uncorrelation and unit variance of white noise errors w_t, the mean square error (MSE) of $y_{\ell,t}$ is:

$$\mathrm{MSE}[y_{\ell,t}] = \mathbb{E}(y_{\ell,t+f} - y_{\ell,t})^2 = \sum_{n=1}^{N} (\theta^2_{\ell n,0} + \cdots + \theta^2_{\ell n,f-1}). \tag{10}$$

Now, $(\theta^2_{\ell n,0} + \cdots + \theta^2_{\ell n,f-1}) = \sum_{i=0}^{f-1} (e'_\ell \Theta_i e_n)^2$ can be interpreted as the contribution of innovations in variable n to the MSE (or the forecast error variance) of the f-step forecast of variable ℓ, where e_n is the n-th column of I_n.

Finally, the amount of forecast error variance of variable ℓ considering the innovations w_{nt} is given by:

$$\alpha_{\ell n,f} = \frac{\sum_{i=0}^{f-1} (e'_\ell \Theta_i e_n)^2}{\mathrm{MSE}[y_{\ell,t}]}. \tag{11}$$

In summary, $\alpha_{\ell n,f}$ represents the proportion of the f-step forecast error variance accounted for by innovation in variable n. Thus, the forecast error variance is *decomposed* into components accounted for by innovations in the various variables.

Panel of Figs. 7 shows the representation of FEVD for each time series belonging to the considered VAR process up to a forecast horizon of 25. For instance, let us consider the first sub-figure reporting the FEVD for the MOS time series. We can say that, in the initial period (up to 2 units of forecast horizon), almost the 100% of variation in the MOS is caused by own shocks (gray bars). Starting from a forecast horizon of 3, we see that variation in the MOS can be also attributed to other variables, and in particular to the round trip delay (green bars). Moreover, starting from a forecast horizon of 10, we can see also the contribution of bandwidth consumption. Likewise, as concerns the FEVD for the BW time series, we can deduce that the variation of BW is caused by shock applied onto BW itself (red bars) for almost the whole forecast horizon range. Forecast error variance of RTD is accounted for by own innovations in the initial period, whereas is influenced by DJB and BW in long runs. BW influences in long runs also the forecast error variance of DJB. Forecast error variance of JIT is accounted for by own innovation for about 50% during the forecast horizon interval, whereas variation on SNR is mainly accounted for by its own innovation during the whole interval.

The following considerations can be drawn from the FEVD analysis. While BW innovations significantly contribute to the forecast error variance of BW itself for the whole forecast horizon, they contribute to the forecast error variance of MOS, RTD, DJB especially in long runs. It means that we observe a kind of propagating effect of BW on other variables, whose impact is visible only after a number of periods. Conversely, for any forecast horizon, SNR innovations contribute less than 1% to forecast error variance of other variables (except for SNR itself). A similar reasoning holds true for MOS, even if a bigger contribution in innovation is given for JIT and DJB. Such a relationship is even confirmed by the correlation matrix in Fig. 6 giving us an instantaneous snapshot of the relationship among MOS, JIT, and DJB. Finally, the innovation relationship is not immediately visible for the three delay-based time series (RTD, JIT, and DJB) probably due to the high randomness governing such variables. In contrast, we have seen that a "static" relationship is well highlighted in the previous correlation analysis. It is useful to stress that, even if the correlation and FEVD analyses provide two complementary views about the time series behaviors, such analyses are derived in different manners. Correlation analysis, in fact, is used to highlight possible common behaviors underlying the time series (e.g. the Pearson coefficient close to 1 indicates that two time series will increase together), and is performed before the VAR model is built. In other words, such an analysis may not provide immediately interpretable information in case we are interested in getting some insights in the forecasting process, since the VAR model is not build yet at this stage. In contrast, the FEVD analysis is performed after the VAR model is built, thus, it may highlight some long-run interactions among variables not captured in the correlation analysis. It is the case of BW and MOS that exhibit a very scarce correlation (see Fig. 6) but a not negligible long-run interaction (see the FEVD for MOS sub-figure in Panel 7).

5. Forecasting with deep learning techniques

At this point, we have built a multivariate time series model where, through specific analyses, we were able to highlight some subtle relationships among involved time series. Before feeding the deep learning schemes with the estimated multivariate model, we need to perform some data manipulation. Preliminarily, we normalize the time series scale in order to standardize the forecasting procedure. Precisely, we have used a min–max normalization in order to bound time series values into the $[0, 1]$ range:

$$y_{t(norm)} = \frac{y_t - y_{min}}{y_{max} - y_{min}}. \tag{12}$$

Then, we re-frame the estimated multivariate time series into a supervised learning structure by adopting the classic 70/30 rule, meaning that 70% of data is for training and 30% for test. Such a re-framing is obtained by using a sliding window (containing 70/30 of training/test samples) which moves one-step across the time [50,51]. In this way, at each time step the window contains more and more updated training/test data. This reframing procedure is well represented in Fig. 8, where a sliding window (dashed red rectangle) moves forward as time elapses, leaving behind past samples (in pale red) and acquiring new samples (in pale green). We note that, with this procedure, all the time series are captured in a given time interval (the window), and then they feed the various deep learning strategies to perform the forecasting.

In the rightmost part of Fig. 2 we have reported the three deep learning schemes exploited to face the forecasting problem. In the following, we provide some details about the adopted deep learning techniques, whose main hyperparameters have been summarized in Table 1.

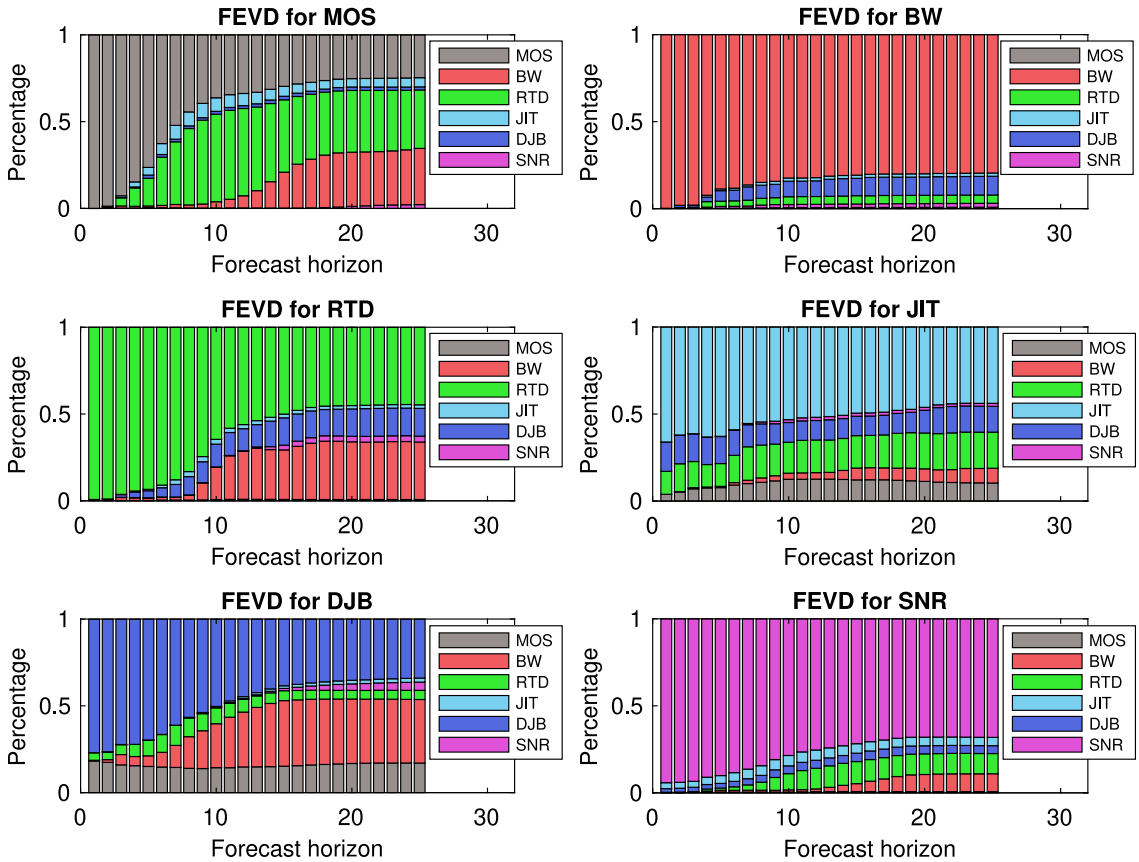

Fig. 7. Forward Error Variance Decomposition (FEVD) for each time series.

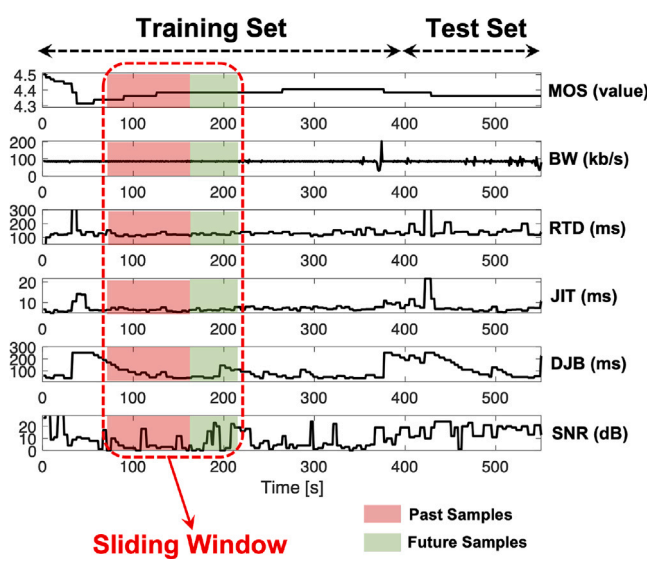

Fig. 8. The sliding window technique: time series reframed as a supervised learning problem.

5.1. Forecasting with CNNs

As the name suggests, the operation which characterizes CNNs is the *convolution* [52]. It can be seen as a sliding filter applied to the time series in order to retain the most important information aimed at improving the quality of forecasting. When applied to the image domain (the most common CNNs application) such a filter has two dimensions (width and height). If applied to the time series domain, the filter is supposed to operate into one dimension (the time), and performs a transformation of the time series by moving forward along the temporal direction [53]. Pragmatically, the output of the *convolutional layer* is a filtered time series which includes the most representative features. Such an output feeds a *flatten layer* which is in charge of condensing the filtered time series into a single one-dimensional vector. Finally, a *dense layer* will interpret the results of convolution and provides the desired forecasting. A schematic representation of the 3-layer CNN employed here is provided in Fig. 2 (top-right part).

5.2. Forecasting with LSTM

LSTM [55] represents a particular class of RNNs, where the inputs have a mutual feedback relation. This peculiarity makes such structures suitable to represent data characterized by sequences, including: DNA sequences, written text (intended as a sequence of words), time series. Standard RNNs suffer from the so-called *vanishing gradient* problem related to the difficulty in managing long-range dependencies in presence of too long sequences. LSTM solves such a problem introducing a particular unit (or cell), whose structure (depicted in the rightmost part of Fig. 2 - middle panel) allows to store the result of past information. Each cell has its own *state* which regulates the information flow through 3 *gates*: the *input* gate (included in a dashed red shape and indicated by I in the LSTM unit of Fig. 2) decides which information must be retained and encoded into the cell state; the *forget* gate (included in a dashed red shape and indicated by F in the LSTM unit of Fig. 2) decides which information must be ignored as new data enters the network; the *output* gate (included in a dashed red shape and indicated by O in the LSTM unit of Fig. 2) decides which information can be transferred to the next LSTM unit. Within a single unit, two *activation functions* in charge of manipulating the information are present: the Sigmoid (σ)

Table 1
Adopted hyperparameters for the exploited deep-based techniques.

Technique	Hyperparameters
CNN	· Stochastic Gradient Descent (Adam version - [54]) · Learning rate = 0.01 · Training loss function: MSE · Number of CNN filters=50 · Kernel size = 6 · Dense layer size = 6 · Dropout layer with dropout rate = 0.5 · Epochs = 100 · Input shape = (12,6)
LSTM	· Stochastic Gradient Descent (Adam version) · Learning rate = 0.01 · Training loss function: MSE · Number of LSTM units=50 · Dense layer size = 6 · Dropout layer with dropout rate = 0.5 · Epochs = 100 · Input shape = (12,6)
GRU	· Stochastic Gradient Descent (Adam version) · Learning rate = 0.01 · Training loss function: MSE · Number of GRU units=50 · Dense layer size = 6 · Dropout layer with dropout rate = 0.5 · Epochs = 100 · Input shape = (12,6)

which maps the information values in the range $(0, 1)$ and defined as

$$\sigma(x) = \frac{1}{1 + e^{-x}}, \tag{13}$$

and the Hyperbolic Tangent (*tanh*) which maps the information values in the range $(-1, 1)$. The three main equations describing analytically the output of forget, input, and output gates, are respectively:

$$f_t = \sigma(w_f[h_{t-1}, y_t] + b_f), \tag{14}$$

$$i_t = \sigma(w_i[h_{t-1}, y_t] + b_i), \tag{15}$$

$$o_t = \sigma(w_o[h_{t-1}, y_t] + b_o), \tag{16}$$

where: h_{t-1} is the input vector result of the hidden layer, $w_{f,i,o}$ are the weights matrices and $b_{f,i,o}$ the bias vectors for forget, input, and output gates, respectively.

5.3. Forecasting with GRU

GRU [56] belongs to the RNNs family and it can be considered as an evolution of LSTM, with the ability of solving the vanishing gradient problem relying on a lighter cell structure. Differently from LSTM, in fact, GRU has two gates in place of three: the *reset* gate (included in a dashed red shape and indicated by R in the GRU unit of Fig. 2) determines the amount of past information to forget; the *update* gate (included in a dashed red shape and indicated by U in the GRU unit of Fig. 2) plays a role similar to the LSTM forget and input gates since it determines which information to keep and which one to ignore. Also GRU units contain both σ and *tanh* activation functions. Even if performance is quite similar to LSTM, GRU networks are expected to be a bit faster due to the simplified internal cell architecture. The two main equations describing analytically the output of reset and update gates, are respectively:

$$r_t = \sigma(w_r[h_{t-1}, y_t] + b_r), \tag{17}$$

$$z_t = \sigma(w_z[h_{t-1}, y_t] + b_z), \tag{18}$$

where: h_{t-1} is the input vector result of the hidden layer, $w_{r,z}$ are the weights matrices and $b_{r,z}$ the bias vectors for reset and output gates, respectively.

Please note that, for visualization comfort, in Fig. 2 we have reported the internal structure of single LSTM and GRU units. But, similarly to the CNNs, the complete LSTM and GRU networks are realized by combining multiple LSTM/GRU units with a final dense layer (see Table 1 for details).

The order of magnitude for adopted hyperparameters of deep-based techniques have been derived from technical literature [57] and chosen all similar in order to achieve meaningful comparative results (see again Table 1).

6. Experimental results

Before delving into the experimental part, it is useful to summarize the fundamental steps taken up to this point and synthetically represented in Fig. 2. Once VoIP traffic has been collected in a real environment and meaningful network metrics extracted and organized as time series, we constructed a multivariate time series model VAR(p). This model consists of a linear combination of the past values of all involved time series with lag value p estimated using the AIC criterion (see Section 4.1). The obtained VAR model was then employed to investigate specific statistical properties of the time series (e.g., uncorrelation of residuals, stationarity), and to gain better insights into their relationships (see FEVD in Section 4.2). Subsequently, the model output y_t was manipulated by reframing its temporal structure into a supervised learning format through the sliding window mechanism (see Section 5). Finally, the manipulated data were normalized and then used as input of three deep learning schemes (CNN, LSTM, GRU) which yield the final temporal forecasting output. Obviously, the more precise the forecast result, the more suitable the employed technique. It is now useful to visualize the behavior of collected metrics and the corresponding forecasts.

Fig. 9 includes six panels reporting the time series representative of the considered network metrics[2]: MOS, BW, RTD, JIT, DJB, and SNR. In each panel, the black curve represents the ground truth, namely the measured trend of the gathered metric. For visualization comfort, we have limited the observation period between 350 and 400 seconds. Moreover, from 400 to 550 seconds, we have the *Prediction Zone* which contains the results of all employed forecasting techniques. The forecasting techniques have been grouped into three parts: first, we show VAR (in green) as representative of the pure statistical strategy; then we show CNN (in orange), LSTM (in blue), GRU (in violet) representing the pure deep-based strategies; finally, we show the proposed hybrid strategies, namely VAR-CNN (in gray), VAR-LSTM (in turquoise), VAR-GRU (in red).

Let us start analyzing the MOS panel. First, we note that the MOS time series exhibits a sort of step-like behavior due to the scarce granularity of such a metric which does not allow to capture subtle oscillations. Moreover, we can interestingly note that, on average, the perceived quality of the call pattern is reasonably good, since the actual MOS value never drops below 4.3 (recall that 5 encodes the highest perceived quality). If we move into the forecasting zone, we can roughly observe the ability of each technique to follow the actual metric value. In the performance figure further ahead, we will quantify precisely the efficiency of each forecasting technique. This notwithstanding, at a first sight it is easy to recognize how the VAR technique has a difficulty in fitting the actual MOS values especially in the first part of forecasting (approximately between 400 and 430 seconds), where a descending peak around 4.1 value is visible. Such a behavior is basically due to the fact that VAR suffers from the irregularities affecting real time series, although the lag is the optimal one.

By analyzing the BW panel, we can see that, except for occasional peaks, this metric fluctuates around a value of 90 kb/s. Since such peaks reflect a non-linear behavior of the BW time series, it is immediate to

[2] Our collected datasets along with details about a Python routine implementing our forecasting hybrid strategies can be found at: https://github.com/mariodim/Hybrid-Learning-Strategies/

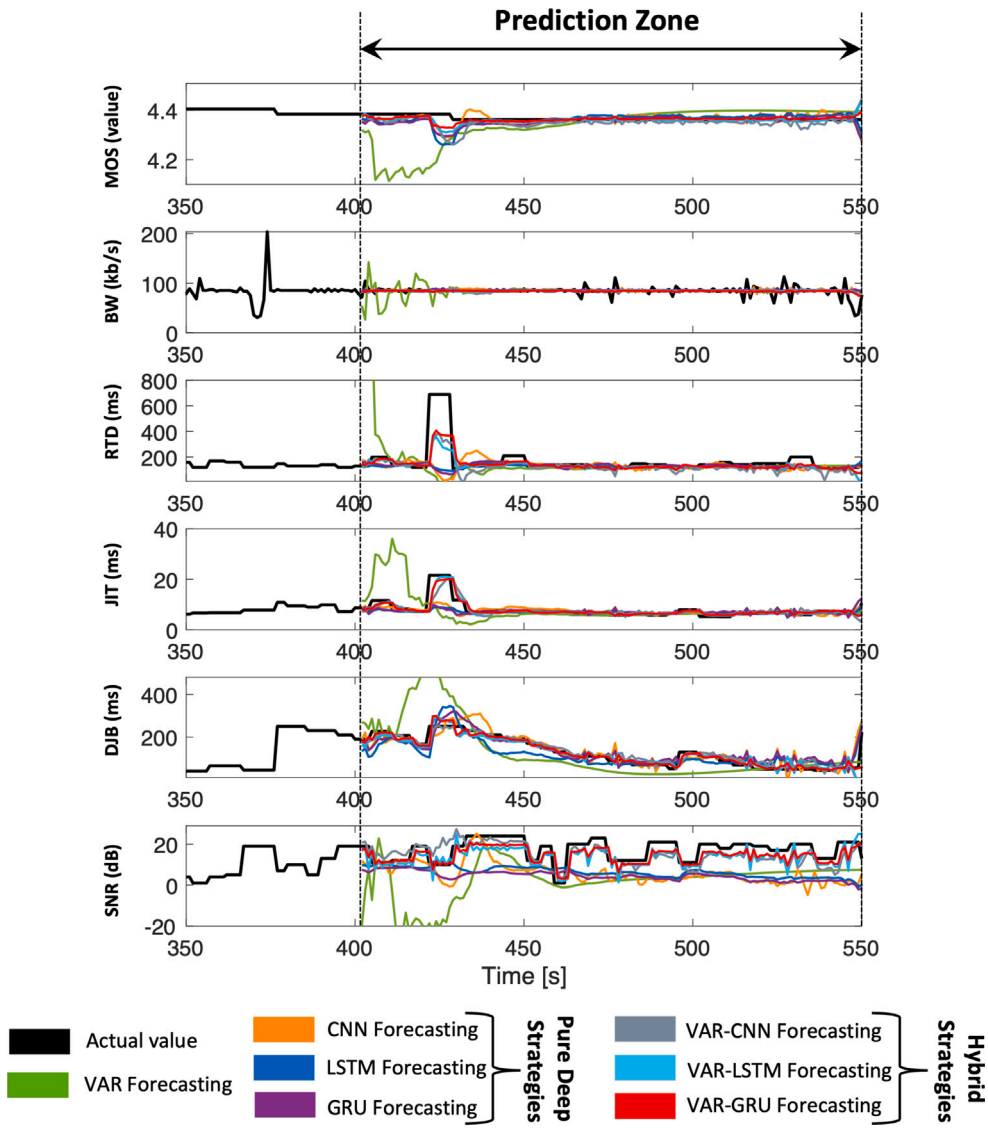

Fig. 9. Multivariate time series forecasting of cellular traffic features via: VAR, *Pure Deep* strategies (CNN, LSTM, GRU) and *Hybrid* strategies (VAR-CNN, VAR-LSTM, VAR-GRU).

see that VAR model fails to capture some high irregularities. Among the remaining techniques (pure and mixed) a satisfactory fitting with the original value may be derived.

Let us now move to the RTD panel. In line with the values of mobile networks (hundreds of milliseconds [58]), also in our case the RTD oscillates between 100 and 150 ms, approximately (in contrast, typical values in fixed networks are around tens of milliseconds). Interestingly, in the RTD panel, we can notice an unusual peak located at about $t = 430$ sec (corresponding to an RTD peak value of 670 ms), which is presumably due to temporary network congestions. We can also notice how such a condition reflects a little degradation in the voice quality, whose MOS values decrease in correspondence with the observed RTD peak. Differently from the MOS time series, RTD exhibits a greater variability, thus forecasting becomes more challenging. From a rough visual inspection, we can see that VAR, CNN, and VAR-CNN are not always able to follow the actual RTD value. In contrast, VAR-LSTM and VAR-GRU appear to be able to track also the unusual peak at $t = 430$ sec.

As regards JIT, we can first see how such a value lies under 10 msec most of the time. Such a value is perfectly in line with telco standards [58] whose average value is prescribed to be targeted at less than 30 ms. This is the reason why the jitter exhibits low fluctuations. Probably, as in the case of RTD, the unusual peak at $t = 430$ sec is due to some temporary network irregularity. Also in this case we can appreciate how hybrid techniques (in particular, VAR-LSTM and VAR-GRU) are capable of well fitting the underlying original time series.

Some regularities are exhibited by DJB, characterized by up-down repeating patterns. It is useful to recall that, even though such a metric is expressed in milliseconds, it is associated with the number of packets employed by the de-jittering buffer. Typically, a dedicated algorithm is in charge of allocating/de-allocating the packets by continuously estimating the jitter across the communication channel. As regards the forecasting ability, a visual inspection reveals how the best are VAR-LSTM and VAR-GRU which allow to follow the up-down trend thus producing satisfying forecasting results.

Finally, the SNR time series shows many irregularities in line with the nature of the indicator itself which includes two measures: the signal (intended as the voice) and the channel noise. The SNR oscillates between 0 (meaning that signal and noise have both the same power)

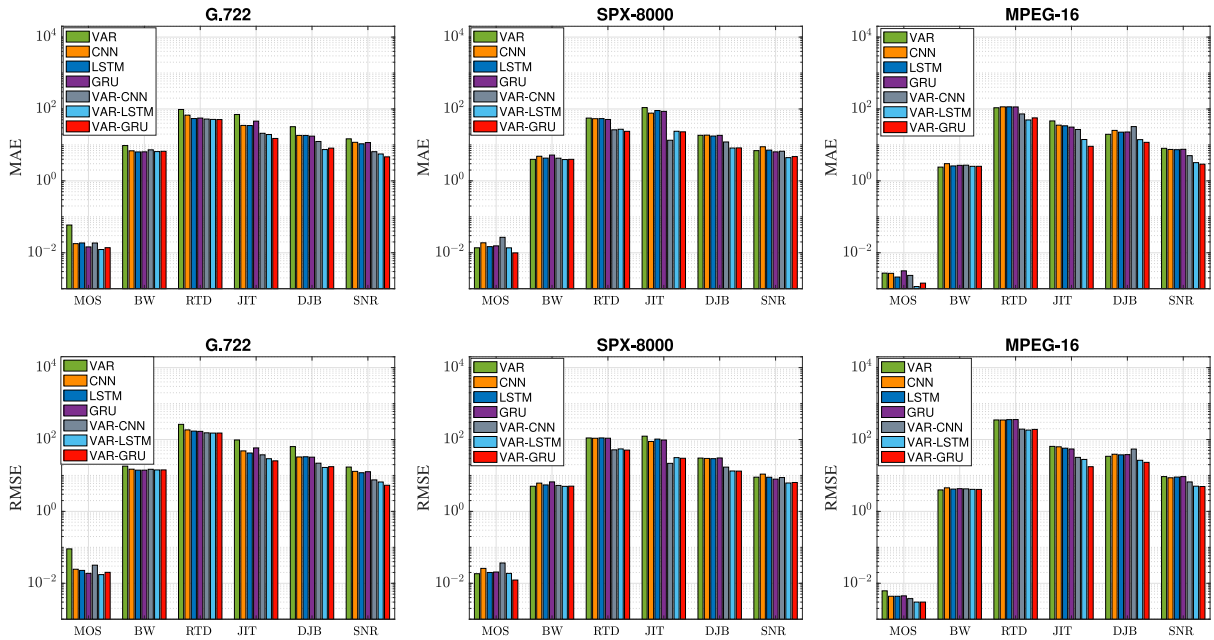

Fig. 10. MAE values (uppermost figures) and RMSE values (lowermost figures) grouped per codec (G.722, SPX-8000, MPEG-16) and per metric (MOS, BW, RTD, JIT, DJB, SNR).

and 20 dB. According to standard prescriptions [59], a good SNR for voice communication should be in the order of 20 dB but, in our case, the extreme variability of the mobile environment causes degradation of such a metric. We note that, although SNR does not exhibit excellent values, the perceived voice quality is good (MOS > 4.3). This apparently counterintuitive relation is justified by the presence of software tools such as dynamic codecs or noise cancelers which allow to make intelligible a conversation even if in presence of bad channel conditions. As regards the forecasting quality, we can see that VAR, CNN, LSTM, and GRU produce forecasting results located in an wide interval of dBs, whereas actual value oscillates between 10 and 20 dB, on average. Conversely, all hybrid strategies are able to well track the SNR actual value.

6.1. Performance discussion

Even if a visual inspection is useful to have a coarse idea of the forecasting ability of each technique, in this section we quantify such an ability in terms of: (i) forecasting error associated to each strategy, and (ii) required computational time. The forecasting error is accounted for by two popular indicators, the test Mean Absolute value of Errors (MAE) and the test Root Mean Square Error (RMSE):

$$MAE_i = \frac{\sum_{t=1}^{L} |y_{it} - \hat{y}_{it}|}{L}, \tag{19}$$

$$RMSE_i = \sqrt{\frac{\sum_{t=1}^{L} (y_{it} - \hat{y}_{it})^2}{L}}, \tag{20}$$

computed for each time series $i = 1, \ldots, 6$, where $\hat{y}$ represents the forecasted value and L the time series length. The reason for considering both performance indicators is because MAE weights in an equal manner the individual differences by applying a linear rule, whereas RMSE gives a relatively high weight to large errors by applying a quadratic rule.

We want to recall that, all the analyses presented in the previous sections are based on a 50,000 packets voice flow exploiting the G.722 audio codec.

Since the usage of a specific codec influences the behavior of the voice flow (and, then, of corresponding QoE/QoS metrics), we consider also 2 new codecs, namely SPX-8000 (8 kb/s of bit rate and 8 KHz of sampling rate) and MPEG-16 (16 kb/s of bit rate and 16 KHz of sampling rate). Similarly for the G.722 case, also such two new collected voice flows contain an approximate amount of 50,000 voice packets. Also for such new experiments we have built a VAR model, and estimated the optimal lags amounting to 8 for the SPX-8000 voice flow, and to 11 for the MPEG-16 voice flow. This notwithstanding, the whole set of analyses (e.g., autocorrelation of residuals, FEVD, etc.) for these latter 2 experiments has been omitted for space constraints, thus we prefer to focus on the final performance for a significant comparison. Accordingly, Panel 10 reports: MAE values for G.722, SPX-8000, and MPEG-16 (uppermost figures), and RMSE values for G.722, SPX-8000, and MPEG-16 (lowermost figures). Such values have been expressed onto a logarithmic scale in order to better appreciate the different orders of magnitude.

A common behavior that can be easily recognized is that MAE/RMSE values associated to hybrid strategies (VAR-CNN, VAR-LSTM, VAR-GRU) are, on average, lower than MAE/RMSE values pertaining to pure strategies (VAR, CNN, LSTM, GRU), obviously implying better performance of hybrid strategies.

As regards to MOS, we see that MAE/RMSE values are substantially smaller than values corresponding to the remaining metrics. The reason is that, differently from other metrics, MOS time series varies slowly and in a limited range of values, thus the forecasting strategies have the possibility to track the actual values in a more accurate manner.

For the remaining metrics, MAE/RMSE values are definitely higher due the presence of more irregularities, translating in a more challenging forecasting. Among the pure strategies, VAR (green bars) seems to exhibit the worst efficiency (namely, the highest MAE/RMSE value) in the majority of cases due to the presence of high irregularities which do not allow to perfectly characterize a time series in terms of simple linear combinations of past values. Indeed, among the pure deep learning techniques, we can observe that CNN (orange bars), LSTM (blue bars), and GRU (violet bars) exhibit similar performance, since their internal structures (convolutional for CNN and gate-based for LSTM and GRU) allow to efficiently manage time-variant information.

Finally, among the proposed hybrid strategies, VAR-GRU seems to exhibit the best performance, or, in some cases (e.g., RTD, DJB), at the same level of VAR-LSTM. For example, in the case of MOS (G.722 codec), the MAE associated to the VAR-GRU technique (amounting to 0.0138) decreases of about 76% w.r.t. the MAE associated to VAR (amounting to 0.0590) and of about 5% w.r.t. the MAE associated to GRU (amounting to 0.0145).

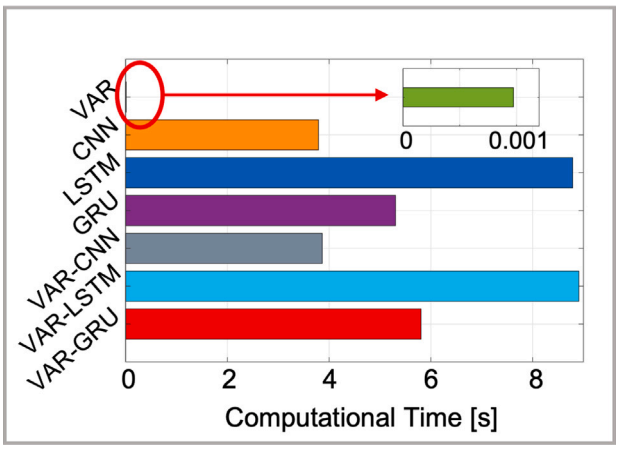

Fig. 11. Computational time for each technique.

6.2. Computational time analysis

The forecasting error analysis is complemented by a temporal performance analysis whose results are shown in Fig. 11. Here we report the computational time required by each strategy averaged on the three experiments (G.722, SPX-8000, and MPEG-16), and using a general-purpose PC (2 virtual CPUs 2.30 GHz and 13 GB of RAM). The first thing catching the eye is the extreme processing speed of VAR amounting approximately to 0.001 s This is justified by the fact that constructing a multivariate time series through linear combinations does not require a huge computational effort. In contrast, deep learning schemes typically rely on complex structures which make the forecasting task more time consuming. In particular, the highest computational time is exhibited by LSTM. Due to the lighter structure made of 2 gates in place of 3, GRU is faster than LSTM and a bit slower than CNN. Interestingly, the hybrid strategies are characterized, approximately, by the same computational times of the corresponding pure deep-based strategies, since the preliminary operation introduced by VAR is negligible in terms of time consumption.

In summary, we can conclude that: (*i*) the smart combination of VAR strategies with deep learning techniques is advantageous in terms of error reduction w.r.t. corresponding pure strategies, with negligible price to pay in terms of extra computational time; (*ii*) among the hybrid learning strategies proposed, the best one is the VAR-GRU in terms of forecasting error/time consumed trade-off. The reason is that GRU model has a flexible cell structure which allows to regulate the sequence flow (e.g., by selectively retaining or discarding some information) across the time, thus it is particularly suited to cope with time series data which, in turn, are expressed as the optimal combination of network metrics produced by VAR.

7. Conclusion and future prospects

In this work we have advanced three hybrid strategies (VAR-CNN, VAR-LSTM, VAR-GRU) which combine statistical and deep learning concepts aimed at forecasting the temporal behavior of crucial mobile network metrics. The purpose is to provide useful analysis tools to mobile network operators aimed at optimizing network resources. Our framework relies on a pipeline which combines the *pros* of two approaches: (*i*) a Vector Autoregressive (VAR) model, a statistical method employed to encode all the metrics within a multivariate time series structure which takes into account the mutual interdependency among the metrics, (*ii*) three deep-based techniques (CNN, LSTM, GRU) which are able to learn the behavior of such a multivariate model and to produce accurate forecasts of future values. Performance results applied to real-environment network metrics reveal the effectiveness of the proposed hybrid strategies with respect to their pure counterparts. This work can be expanded across several directions: first, the experiments can be extended in rural and/or highly dense areas in order to analyze the environmental effects on the metrics, or by considering a different type of data traffic (e.g., video). Then, it could be useful to trace the location of the mobile phone on the vehicle so as to gain some relations between specific positions (e.g., in a tunnel) and metric behaviors. Finally, it could be interesting to make comparison against classic univariate strategies to quantify the differences with the multivariate approach adopted in this work.

CRediT authorship contribution statement

Mario Di Mauro: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Giovanni Galatro:** Software. **Fabio Postiglione:** Writing – review & editing, Formal analysis. **Wei Song:** Writing – review & editing. **Antonio Liotta:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data gathered from experiments (along with routines implementing the analyzed forecasting strategies) are available through github platform as indicated into the paper.

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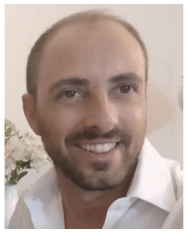

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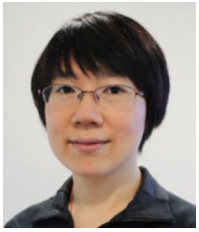

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